

MA 521: Functional Analysis

Course Code	MA 521
Course Name	Functional Analysis
L-T-P-C	3-1-0-4
Prerequisites	MA-511 (Real Analysis)
Intended for	M.Sc./B. Tech./M.S./Ph.D.
Distribution	Core for M.Sc. in applied Mathematics and Elective for other discipline
Approval	10th Senate (Jun 2016)

Course Contents

- **Module I** [5 Lectures]
Normed spaces, Examples of Normed Spaces, Subspaces of Normed Spaces, Quotient Normed Spaces, Riesz Lemma, Finite-Dimensional Normed Spaces, Convex Subsets of Normed Spaces, Stronger and Equivalent Norms, Strictly Convex Normed Spaces.
- **Module II** [5 Lectures]
Linear Maps Between Normed Spaces, Continuity of linear maps, Examples of Discontinuous Linear Maps on Infinite Dimensional Normed Spaces, Various Criterion for Continuity of Linear Maps, Linear Functionals, Examples of Continuous Linear Maps, Necessary Conditions for the Continuity of Transformations defined by Infinite Matrices, Operator Norm of Bounded Linear Maps, Operator Norm of Transformations defined by Finite Matrices.
- **Module III** [5 Lectures]
Hahn-Banach Separation Theorem, Hahn-Banach Extension Theorem, Consequences of Hahn-Banach Extension Theorem, Uniqueness of the Hahn-Banach Extension, Banach Limits.
- **Module IV** [8 Lectures]
Banach Spaces, Subspaces of Banach Spaces, Quotient Banach Spaces, Product of Banach Spaces, Canonical Embedding of Normed Spaces, Schauder Basis, Uniform Bounded Principle and its Applications, Banach-Steinhaus Theorem..
- **Module V** [6 Lectures]
Closed Maps, Closed graph theorem, Linear Projections, Open Maps, Quotient Maps, Open Mapping Theorem and its Applications, Bounded Inverse Theorem.
- **Module VI** [5 Lectures]
Spectrum of Bounded Operators, Resolvent Set, Eigen-spectrum, Approximate Eigen- spectrum, Spectrum of the Right Shift Operator, Compact Operators on Normed Spaces, Spectrum of Compact Operators.
- **Module VII** [8 Lectures]
Inner Product Spaces, Orthonormal Sets, Bessel's Inequality, Riesz-Fischer Theorem, Fourier Expansion, Parseval Formula, Projection and Riesz Representation Theorems, Bounded Operators and Adjoints, Normal, Unitary and Self-Adjoint Operators.

Text Books

1. B.V. Limaye, **Functional Analysis**, Revised 3rd Edition, New Age International Private Limited, 2017.
2. B.V. Limaye, **Linear Functional Analysis for Scientists and Engineers**, Springer, Singapore, 2016.
3. J.B. Conway, **A Course in Functional Analysis**, 2nd Edition, Springer, 1990.

References

1. E. Kreyzig, **Introductory Functional Analysis with Applications**, John Wiley & Sons, New York, 1989.
2. K. Yoshida, **Functional Analysis**, 6th Edition, Springer, 1995.
3. C. Goffman and G. Pedrick, **A First Course in Functional Analysis**, Prentice-Hall, 1974.
4. A. Taylor and D. Lay, **Introduction to Functional Analysis**, Wiley, New York, 1980.